

MATH 211
Lab Problems - Week 10

1. Compute $\left(-\frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}}\right)^8$.

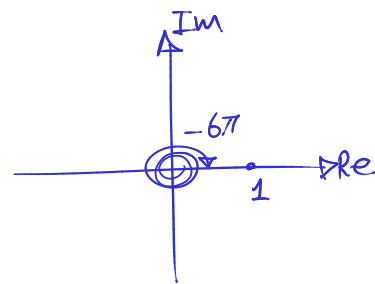
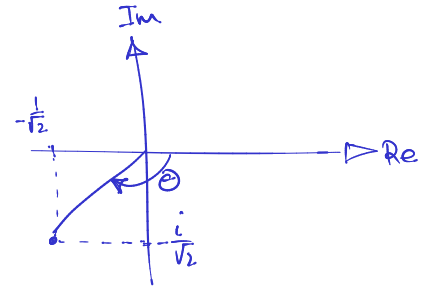
$$r = \sqrt{\frac{1}{2} + \frac{1}{2}} = 1$$

$$\theta = -\frac{3\pi}{4}$$

De Moivre's thm

$$\left(-\frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}}\right)^8 = \left(1 \cdot e^{i \cdot \left(-\frac{3\pi}{4}\right)}\right)^8 = 1 \cdot e^{i \cdot (-6\pi)}$$

$$\left(= \cos(-6\pi) + i \sin(-6\pi)\right) = 1$$



2. Find all roots of $(\sqrt{3} - i)$.
the 3rd

$$|\sqrt{3} - i| = \sqrt{3+1} = 2$$

Let $q = re^{i\theta}$ such that $q^3 = \sqrt{3} - i = 2e^{i(-\pi/6 + 2k\pi)}$, $k \in \mathbb{Z}$

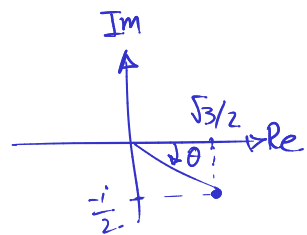
Then $q^3 = r^3 e^{i3\theta} = 2e^{i(-\pi/6 + 2k\pi)}$

$$r = \sqrt[3]{2}$$

$$3\theta = -\frac{\pi}{6} + 2k\pi, k \in \mathbb{Z}$$

$$\Rightarrow \theta = \frac{-\pi/6 + 2k\pi}{3}, k \in \mathbb{Z} \rightarrow \begin{cases} k=0 \Rightarrow \theta = -\frac{\pi}{18} \\ k=1 \Rightarrow \theta = \frac{11\pi}{18} \\ k=2 \Rightarrow \theta = \frac{23\pi}{18} \end{cases}$$

$$\Rightarrow q = \begin{cases} \sqrt[3]{2} e^{i(-\pi/18)} \\ \sqrt[3]{2} e^{i(11\pi/18)} \\ \sqrt[3]{2} e^{i(23\pi/18)} \end{cases}$$



$$\theta = -\frac{\pi}{6} + 2k\pi$$

$$k \in \mathbb{Z}$$

(why is it enough to let $k=0,1,2$, only?)

3. Solve $x^2 + 2x + (1+i) = 0$.

By quadratic formula: $x = \frac{-2 \pm \sqrt{4 - 4 \cdot 1 \cdot (1+i)}}{2} = \frac{-2 \pm \sqrt{-4i}}{2} = \frac{-2 \pm 2\sqrt{-i}}{2} = -1 \pm \sqrt{-i}$ (*)

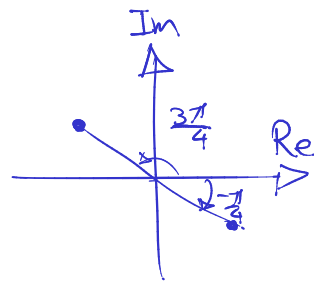
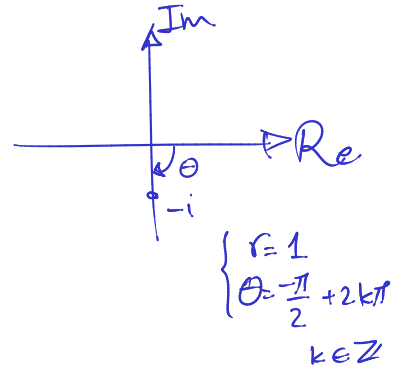
In order to find second roots of $-i$:

$$q = re^{i\theta} \rightarrow q^2 = r^2 e^{i2\theta} = 1 \cdot e^{i(-\frac{\pi}{2} + 2k\pi)}, k \in \mathbb{Z}$$

$r=1$ $\theta = -\frac{\pi}{4} + k\pi, k \in \mathbb{Z}$

$$\begin{cases} \theta = -\frac{\pi}{4} \\ \theta = \frac{3\pi}{4} \end{cases}$$

$$\rightarrow q = \begin{cases} 1 \cdot e^{i(-\frac{\pi}{4})} \\ 1 \cdot e^{i\frac{3\pi}{4}} \end{cases} \rightarrow q = \begin{cases} \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i \\ -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i \end{cases} = \pm \frac{\sqrt{2}}{2} (1-i)$$



Then by (*) $x = -1 \pm \frac{\sqrt{2}}{2} (1-i) = \begin{cases} -1 + \frac{\sqrt{2}}{2} - i \\ -1 - \frac{\sqrt{2}}{2} + i \end{cases}$

4. Find eigenvalues and eigenvectors of the matrix $\begin{bmatrix} 4 & 0 & 0 \\ -1 & 3 & 5 \\ 1 & 1 & -1 \end{bmatrix} = A$.

$$\det(A - \lambda I) = \det \begin{pmatrix} 4-\lambda & 0 & 0 \\ -1 & 3-\lambda & 5 \\ 1 & 1 & -1-\lambda \end{pmatrix} \xrightarrow{\text{by Laplace expansion over the first row.}} (4-\lambda) \left((3-\lambda)(-1-\lambda) - 5 \right) = (4-\lambda)(\lambda-4)(\lambda+2)$$

e-val's of A are $4, 4, -2$.

Let $\lambda_1 = 4$ (multiplicity 2) $\leadsto A - 4I = \begin{bmatrix} 0 & 0 & 0 \\ -1 & -1 & 5 \\ 1 & 1 & -5 \end{bmatrix} \xrightarrow{\text{rref}} \begin{bmatrix} 1 & 1 & -5 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

and $\lambda_2 = -2$ (multiplicity 1) $\leadsto A + 2I = \begin{bmatrix} 6 & 0 & 0 \\ -1 & 5 & 5 \\ 1 & 1 & 1 \end{bmatrix} \xrightarrow{\text{rref}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

set of all sol'n's: $\left\{ s \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} : s \in \mathbb{R} \right\}$ to $(A + 2I)x = 0$

a set of basic solutions: $\left\{ \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \right\}$ $\leftarrow v_3$
e-vector corresponding to $\lambda_2 = -2$

set of all sol'n's: $\left\{ s \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix} : s, t \in \mathbb{R} \right\}$ to $(A - 4I)x = 0$

a set of basic solutions: $\left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix} \right\}$ $\leftarrow v_1, v_2$
e-vectors corresponding to $\lambda_1 = 4$

Check:

$$Av_1 = \begin{bmatrix} 4 & 0 & 0 \\ -1 & 3 & 5 \\ 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -4 \\ 4 \\ 0 \end{bmatrix} = 4 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \checkmark$$

$$Av_2 = \begin{bmatrix} 4 & 0 & 0 \\ -1 & 3 & 5 \\ 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 20 \\ 0 \\ 4 \end{bmatrix} = 4 \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix} \checkmark$$

$$Av_3 = \begin{bmatrix} 4 & 0 & 0 \\ -1 & 3 & 5 \\ 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \\ 2 \end{bmatrix} = -2 \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \checkmark$$